

UEC404: SIGNALS AND SYSTEMS				
	L	T	P	Cr
	3	1	2	4.5
Course Objective: The aim of this subject is to develop analytical capability of students, by which they would be able to handle real-time signal processing related problems and projects. The knowledge of various transforms will help students to work in multi-disciplinary fields of engineering in group activities.				
Syllabus Representation of Signals and Systems: Signals, Basic Continuous and discrete Time signals and systems, Energy and power signals, System modeling concepts, Linear time invariant systems, Representation of signals in terms of impulses, Discrete time LTI systems continuous time LTI systems, Properties of LTI systems, Systems described by differential and difference equations, Sampling theorem, Quantization. Fourier Analysis: Continuous and discrete time Fourier series, Trigonometric and exponential Fourier series, Properties of Fourier series, Parseval's theorem, Line spectrum, Continuous and discrete time Fourier transforms and its properties, Analysis of discrete time signals and systems, Correlation, Autocorrelation, Relation to Laplace transform. Z-Transform: Definition of Z-transform and Properties of Z-transform, Inverse Z-transform - Power series, partial fraction expansion, residue method and their comparison, Relation between Z.T. and F.T, Transfer function, Discrete time convolution, Stability considerations, Time domain and frequency domain analysis, Solution of difference equation, Applications of Z-transforms. Introduction to Fast Fourier Transforms: Discrete Fourier transform, Properties of DFT, Fast Fourier transforms, Divide and Conquer Approach, Decimation in time and decimation in frequency, Radix-2 FFT, Radix-4 FFT algorithms, Linear Convolution, Circular Convolution, Power spectrum and correlation with FFT. Other transforms: Discrete Sine Transform, Discrete Cosine Transform and its types. Familiarization with Standard: IEEE 1641-2004, IEEE 1139-1999.				
Laboratory Work: Signal generation, Solving difference equation, Calculating Z-transform, Linear and Circular convolution, Correlation, DFT / IDFT, FFT algorithms using MATLAB.				
Course Learning Objectives (CLO) The students will be able to: - Analyze the properties of signals and systems in the continuous-time-domain. - Analyze the properties of signals and systems in the discrete-time-domain. - Analyze the continuous-time signals and systems in the frequency-domain. - Analyze the discrete-time signals and systems in the frequency-domain.				
Text Books - Oppenheim, A.V. and Willsky, A.S., <i>Signal & Systems</i>, Prentice Hall of India (1997). - Kani, A.N. <i>Signals and Systems</i>, McGraw Hill Higher Education, (2011) - Proakis, J.G. and Manolakis, D.G., <i>Digital Signal Processing Principles Algorithm</i>				

& Applications, Prentice Hall, (2007).

Reference Books

2. Lathi, B.P., *Modern Digital and Analog Communication Systems*, Oxford Univ. Press, 1998
3. Papoulis, A., *Probability Random Variables and Stochastic Processes*, McGraw Hill, 2008